

1. Identify repetitive operational tasks (e.g., log analysis) and automate them using Python.

Make sure you have python installed in your local system or incase using ec2 instance respectively.

Step 1 > create `vi.fetch_logs.py` and save

```
import re

# Log file path (change as needed)
LOG_FILE = "/var/log/syslog"

def fetch_logs():
    try:
        with open(LOG_FILE, "r") as file:
            logs = file.readlines()

        # Filter logs for keywords
        filtered_logs = [log for log in logs if
            re.search(r"CRITICAL|FAILED|WARNING", log, re.IGNORECASE)]

        if filtered_logs:
            print("\n **Filtered Logs Found:**\n")
            for log in filtered_logs:
                print(log.strip())
        else:
            print("\n No critical logs found.\n")

    except Exception as e:
        print(f"\n Error reading log file: {e}\n")

# Run the log fetch function
fetch_logs()
```

Step 2 > Run the Script

make it executable:

```
chmod +x fetch_logs.py
```

Run the script:

```
python3 fetch_logs.py
```

1.